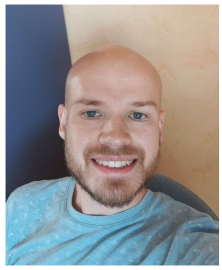
Thomas Jacumin

Born on 5th July, 1995
French citizenship
41 Avenue Georges Clémenceau
06000 Nice
France
+33 6 61 69 78 29
thomas.jacumin@inria.fr



Experiences

September 2023–March 2027 — **Postdoc in Applied Mathematics**, *University Côte d'Azur*, France, under the supervision of Pr. R. Masson

March 2023–March 2025 — **Postdoc in Applied Mathematics**, *University of Lund*, Sweden, under the supervision of Dr. A. Langer

Education

2019–2022 — **PhD in Applied Mathematics**, *University of Haute–Alsace*, France, under the supervision of Pr. Z. Belhachmi

2017–2018 — **Master in Modeling and Applied Mathematics**, *University of Augsburg*, Germany, with high honours

2016–2018 — **Master in Applied Mathematics**, *University of Rouen*, France, with honours

2015–2016 — **1st Year of Engineering School**, *École nationale Supérieure en Informatique, Systèmes Avancés et Réseaux*, Valence, France, validated

2013–2015 — **DEUG of Mathematics "Prépa Écoles Ingénieurs"**, *University of Avignon*, France, with honours

2010–2013 — **Baccalauréat Scientifique**, *Lycée de l'Arc*, Orange, France, with high honours

Publications

- Thomas Jacumin and Andreas Langer. "An Adaptive Finite Difference Method for Total Variation Minimization". In: *Numerical Algorithms* (Mar. 19, 2025). ISSN: 1017-1398, 1572-9265. DOI: `10.1007/s11075-025-02044-6`.
- Zakaria Belhachmi and Thomas Jacumin. "Adjoint Method in PDE-Based Image Compression". In: *Asymptotic Analysis* (Oct. 2024), pp. 1–28. ISSN: 18758576, 09217134. DOI: `10.3233/ASY-241944`.
- Zakaria Belhachmi and Thomas Jacumin. "Optimal Interpolation Data for PDE-Based Compression of Images with Noise". en. In: *Communications in Nonlinear Science and Numerical Simulation* 109 (June 2022), p. 106278. ISSN: 1007-5704. DOI: `10.1016/j.cnsns.2022.106278`.

Preprints

- Zakaria Belhachmi and Thomas Jacumin. *Optimal Transport Model of Optical Flow Estimation: Constant and Varying Illumination Cases*. Dec. 2024.
- Zakaria Belhachmi and Thomas Jacumin. *Iterative Approach to Image Compression with Noise: Optimizing Spatial and Tonal Data*. Sept. 29, 2022. DOI: `10.48550/arXiv.2209.14706`. arXiv: `2209.14706`.

Communications

March 2026 — **Adjoint Method in PDE-Based Image Compression**, *SMAI-MODE 2026*, Nice, France
`https://mode2026.sciencesconf.org/`

March 2026	**Poster – A Coats-Based Formulation for Multiphase Reactive Transport: Application to Hydrogen Storage in Porous Media**, *APPRISE Winter School*, Aussois, France `https://apprise.sciencesconf.org`
February 2026	**PDE-Based Images Compression**, *PhD Seminar*, University Côte d'Azur, France
November 2024	**An Adaptive Finite Difference Method for Total Variation Minimization**, *Seminar*, University of Lund, Sweden
January 2024	**Modèle Transport Optimal pour la Formulation du Flot Optique**, *ANR SOCOT*, University of Haute-Alsace, France `https://codimd.math.cnrs.fr/s/vKWD_JtAu`
July 2023	**Méthodes Adaptatives pour les Problèmes de Discontinuité**, *MAThEOR Days*, University of Strasbourg, France `https://days.matheor.com/2023/`
October 2022	**Mini-symposium – Inverse problems and shape optimization**, *Optimal interpolation data for PDE-based compression of images with noise*, PICOF22 – University of Caen, France `https://picof22.sciencesconf.org/`
June 2022	**PhD Day – "Ma thèse en 180 secondes" format**, University of Haute–Alsace, France
June 2022	**PhD Seminar**, *IRIMAS*, University of Haute–Alsace, France
December 2021	**Poster**, *Calculus of Variations Colloquium*, University of Nancy, France `https://indico.math.cnrs.fr/event/7000/`

Teaching

January–April 2026	**Calculus 2**, *Exercice sessions intended for first-year undergraduate students in Applied Mathematics and Computer Science for the Social Sciences and Mathematics.*, University of Côte d'Azur, France
March–November 2024	**Bachelor Thesis Supervision**, *Adaptive Finite Differences Method*, University of Lund, Sweden
April–May 2022	**Master 1 Project Supervision**, *Image Compression by Image Inpainting*, University of Haute-Alsace, France

Work Groups

November 2022	**Benamou-Brenier Formula**, *Optimal Transport*, University of Haute–Alsace, France `https://juillet.perso.math.cnrs.fr/GTTO.html`
March 2022	**C–transforms**, *Optimal Transport*, University of Haute–Alsace, France

Attended Schools

March 2026	**Winter school APPRISE (Fundamentals of trAnsport Phenomena in PoRous medIa across ScalEs)**, Centre Paul-Langevin, France
July 2023	**Summer school on Numerical Analysis of nonlinear PDEs**, University of Leipzig, Germany

Other Experiences

August 2022	**PhD Seminar Organization**, *Website Development*, IRIMAS, University of Haute–Alsace
June–July 2022	**Colloquium Organization**, *Website Development*, IRIMAS, University of Haute–Alsace `http://www.lmia.uha.fr/colloque-lutz/`
June 2022	**Event Organization**, *Music Festival*, University of Haute–Alsace
September 2021	**AMIES Challenge**, *Trajectories Reconstruction*, Eurecam

Skills

French	Native

English	Good
Japanese	Notions
IT Tools	Python, SageMath, MATLAB, FreeFem++, Latex, C, C++, Java, VHDL, CUDA, Fortran, OpenFOAM

Referees

Belhachmi Z.,	Professor,	IRIMAS,	*University of Haute–Alsace*,	zakaria.belhachmi@uha.fr
Langer A.,	Assistant Professor,	LTH,	*University of Lund*,	andreas.langer@math.lth.se
Masson R.,	Professor,	LJAD,	*University Côte d'Azur*,	roland.masson@univ-cotedazur.fr